CS410 34324 – Lab 1

Gold Star: ADHD Routine Improvement Application

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1. Introduction

Individuals with ADHD diagnoses or ADHD-like traits often struggle with consistent routine development. ADHD is medically classified into three subtypes: Inattentive, Hyperactive-Impulsive, and Combined (Chhabildas et al., 2001, 529). With ADHD comes productivity-altering traits including inattention, hyperactivity, impulsivity, memory problems, and a lack of focus (Staley et al., 2026, p. 890; Advanced Psychiatry Associates, 2026).

ADHD often persists beyond childhood, as do the routine struggles that come with it. Routines may last anywhere from 3 days to "part of a month", and are vulnerable to persistent Routine Adaptations (ADHD Philadelphia, 2026). Routine Adaptations often lead to frustration, anxiety, avoidance, and reduced self-efficacy (Chen et al., 2026, pp. 2, 7). Self-reliance on Routine Adaptations requires tedium creating new schedules while estimating uncertain timeframes. Individuals are also likely subject to error: they may forget to review their schedule, or Task Deviations may occur.

Gold Star is a web-based habit-building and time management platform tailored to those affected by ADHD. It is designed to help users organize their schedules, break down tasks, and build sustainable routines – making it easier to start, stay on track, and follow through every day.

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2. Gold Star Product Description

Gold Star is a web-based productivity and habit-building platform designed primarily for individuals with ADHD struggling with routine creation and maintenance. Many people with ADHD struggle with organization, time management, and keeping up with daily responsibilities. Gold Star gives users a low-stakes environment where tasks, schedules, and habits are monitored without feeling overwhelmed. Gold Star is intended for easier routine development and maintenance, while offering support for users struggling to preserve long-term goal consistency.

2.1. Key Product Features and Capabilities

Gold Star uniquely focuses on helping users organize tasks and schedules through an Adaptive Scheduling system which adjusts to missed tasks and changing priorities. Gold Star evolves with the user, reducing the frustration of falling behind, making it easier to stay on track. It utilizes techniques such as Habit Stacking, Activity Alternation, and the Pomodoro Technique which are recommended for people who struggle with ADHD-like symptoms. Users can choose their productivity level and alter their schedule at any time throughout the day, giving them enough flexibility to feel successful. Users can group tasks by importance level, allowing Gold Star to become personalized to their unique scheduling needs.

Gold Star rewards the user for completing tasks and milestones, giving them a sense of accomplishment. It does not punish them for missing goals, instead featuring a “Roll Over” system, allowing the user to move their missed tasks to another day. Combining flexibility and structure, the platform helps users overcome common ADHD challenges such as task initiation, consistency, commitment, and completion, ultimately improving productivity and reducing daily life stress.

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2.2. Major Components (Hardware/Software)

The system requires no specialized hardware. Users can access the application through any off-the-shelf device with a modern web browser. The server side runs on a single hosting environment supporting the Node.js web server and PostgreSQL database, with deployments being automatically pushed through GitHub. As shown in Figure 1, the system follows a three-tier structure: a Presentation Layer (the web-based UI), an Application Layer (the Node.js web server / API with authentication, and a notification handler), and a Data Layer (PostgreSQL database storing the encrypted user data, tasks, and routine data).

The software to be developed includes the UI/UX built with HTML, CSS, and JavaScript. It will feature a calendar view, routine dashboard, productivity level system, and statistics dashboard. It will also feature a Node.js web server handling routing, logic, authentication components, and a WebSocket-based notification handler. A PostgreSQL database is included to store user account information, saved tasks, routine data, and sensitive user information kept in an encrypted form. Development is supported by off-the-shelf tools: Visual Studio Code, Git, GitHub, GitHub Actions for CI/CD, Jest for testing, and Javadoc for documentation.

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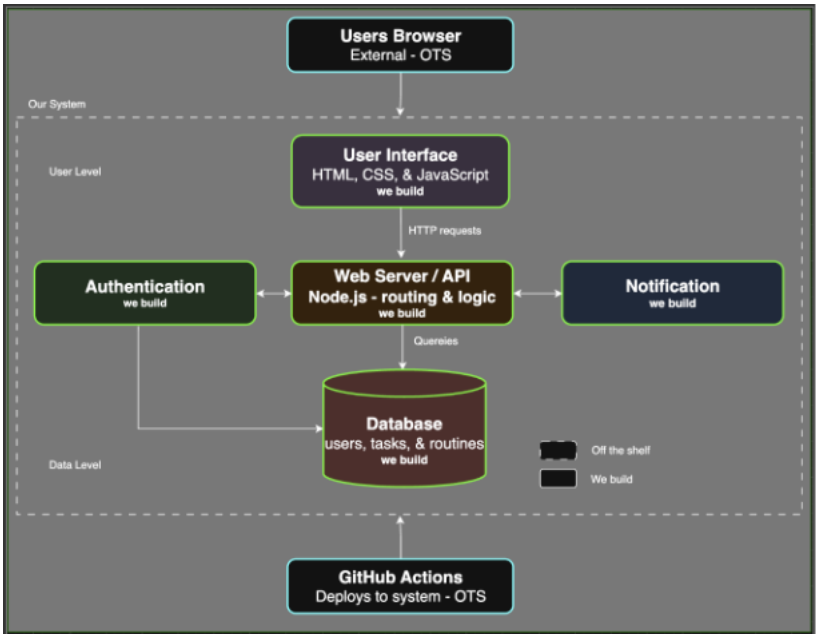

Figure 1: Gold Star's Major Functional Component Diagram.

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3. Identification of Case Study

Individuals with ADHD who struggle with time management, organization, and routine maintenance are Gold Star's primary intended demographic. Many people with ADHD face challenges during schedule creation and consistency due to difficulties with focus, motivation, and efficient task prioritization. To align with Gold Star's main purposes of automating task development, task maintenance, and schedule changes, this consumer demographic serves as the software's primary end user role intending to use the application for improvements in productivity and schedule balancing.

Despite individuals with ADHD serving as Gold Star's primary end user role, this software could also benefit other groups. Additional end user demographics may include parents, guardians, therapists, and other neurodivergent or neurotypical people seeking assistance with improving time management, productivity, and routine building skills. Gold Star was developed for individuals with ADHD, but the application is not intended to exclude users without either ADHD or ADHD-like traits. Regardless of symptoms, people who struggle with organization, productivity, or habit-building consistency are expected to find this application useful for managing daily tasks and stable routines.

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4. Glossary

Activity Alternation – Switching from an activity requiring high mental focus to an activity requiring low mental focus.

Activity Switching – The cognitive process of disengaging from one task and shifting mental focus to another task.

Adaptive Scheduling – To schedule a task which can be completed anytime within a set timeframe. For example, a task which can be completed anywhere inclusively between 7:00 P.M. and 10:00 P.M.

Application Programming Interface (API) - A set of standard rules, protocols, and definitions allowing different software applications to interact with one another.

Attention-Deficit/Hyperactivity Disorder (ADHD) - A common neurodevelopmental disorder that alters the brain's ability to plan, focus, remember instructions, and manage emotions.

Cascading Style Sheets (CSS) - A style sheet language used to determine the visual presentation, layout, and aesthetics of a website.

Git - A digital version control system intended to preserve a history of saved modifications, additions, and deletions to software source code.

GitHub - A cloud-based platform where developers can collaborate on software code.

Gold Star - A web-based habit-building and time management platform designed specifically for individuals with ADHD.

Habit Stacking – Interspersing new habits with existing habits to make task completion easier without feeling overwhelmed.

Hyperactivity - A state of being unusually or abnormally active.

HyperText Markup Language (HTML) - The standard markup language used to create and structure webpages.

JavaScript (JS) - A lightweight programming language used to build dynamic interactivity on web pages.

Jest - An open-source JavaScript testing framework.

JSDoc - An API documentation generator and syntax standard for JavaScript.

Pomodoro Technique – A time-management technique which separates work into 25-minute increments and 5-minute breaks. The standard Pomodoro process repeats this technique 4 times, then a longer break between 20 to 30 minutes is taken before the process repeats.

PostgreSQL - An open-source RDBMS that stores, manages, and retrieves data.

Reinforce - To make something stronger, more effective, or more compelling by adding support.

Relational Database Management System (RBMS) - Software used to store, manage, query, and retrieve data structured in table form.

Rigid Scheduling – To schedule a high-priority task which must either be initiated at or completed before an established time. For example, a task which must start at 6:00 P.M. or must be completed by a deadline of 6:00 P.M.

Routine Adaption – A requirement for an individual to change one's schedule or create new routines.

Routine - A sequence of actions regularly followed, or a fixed program.

Task Deviation – When a task has an unexpected change regarding completion. A task may be accomplished out of order in one's schedule, or it may not be accomplished at all.

Visual Studio Code (VSCode) - A development environment used for software engineering and web development.

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